SPLUNK TIER 1 SOC ANALYST

STUDENT WORKBOOK

Fill-in-the-blank exercises and lab worksheets for Lessons 1–15 and Labs 1–10.

This workbook contains no answers. Complete it during or after each lesson using your own Splunk instance and the BOTS v2 Attack-Only dataset (index=botsv2). Your instructor holds the separate answer key.

Name: Ella Faith M. Sigue **Date:** October 14, 2026

Lesson 1 — The SOC Analyst Mindset & What Splunk Is

VOCABULARY CHECK — fill in the blank

1. A _Security Operations Center_ is a team that watches an organization's systems for signs of attack.
2. A _Tier 1 Security Analyst_ is the first responder who triages alerts and decides what's worth escalating.
3. A(n) _red flag or anomaly flag_ is a system-generated flag that a pattern of events looks worth a human's attention.
4. _Alert Triage_ is the process of quickly sorting alerts by urgency.
5. _Incident Escalation_ is the process of handing a confirmed or high-severity issue to a more senior analyst.

STUDENT EXERCISE

In your own words, write one sentence each defining: log, event, alert, triage, escalation.

A log is a continuous, chronological file that records system data, user activity, or system health over time.

An event is any single, distinct occurrence captured within a network or system, ranging from a successful login to a failed connection.

An alert is an immediate notification triggered when an event violates a predefined security rule or performance threshold.

Triage is the phase where analysts quickly assess, filter, and prioritize alerts to isolate real threats from false positives.

Escalation is the operational process of passing a verified, high-risk incident up to senior specialists for advanced investigation and containment.

Lesson 2 — Installing & Verifying BOTS v2

VOCABULARY CHECK — fill in the blank

1. $SPLUNK_HOME on Linux is _/opt/splunk_.
2. $SPLUNK_HOME on Windows is _C:\Program Files\Splunk_.
3. BOTS v2 Attack-Only must be extracted into the Splunk _etc/apps_ directory.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 _earliest_=0
| stats _count by sourcetype_
```

STUDENT EXERCISE

Run your completed search above. Record the total event count you see, and list the top 3 sourcetypes by count.

68,870,348 events

Top 3 sourcetypes

- access_combined
- activedirectory
- apache_error

Lesson 3 — Reading a Single Log Like an Analyst

VOCABULARY CHECK — fill in the blank

1. _time is the parsed timestamp field Splunk assigns to every event.
2. _raw contains the complete, unmodified original text of an event.
3. A(n) _field_ is a searchable name/value pair extracted from event data.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 earliest=0
| _head_ 5
```

STUDENT EXERCISE

Expand one event fully. Answer: What happened? When? Which host? Any user, IP, or process visible? Anything unusual?

1. What happened?

A client sent an HTTP PUT request to apply the coupon code 20twitter to a Magento 2 shopping cart. The request failed because the coupon code was invalid, and the server returned HTTP 404 Not Found.

2. When did it happen?

August 31, 2017, at 11:17:11.304 AM (displayed log time). The UTC timestamp is 2017-08-31T03:17:11.304235Z.

3. Which host?

The destination host is store.froth.ly, running Apache 2.4.7 on Ubuntu. The destination IP is 172.31.7.2

4. Any user, IP, or process visible?

- Source IP: 211.83.203.115
- Destination IP: 172.31.7.2
- User: No identifiable username is visible.
- Process/application: Magento 2's CouponManagement->set() function handled the coupon request.
- User agent: Chrome 59 on Windows 10 (reported as Windows NT 6.333).

5. Anything unusual?

The coupon application failed, and the response exposed a detailed PHP stack trace, revealing internal Magento application paths and function names. This could be an information disclosure risk. The request came from an external IP, but this single event alone does not prove malicious activity.

Lesson 4 — SPL Fundamentals I — Filtering & Building Blocks

VOCABULARY CHECK — fill in the blank

1. In SPL, a bare space between two search terms means _AND_ (both must be true).
2. The keyword _OR_ means at least one of the grouped conditions must be true.
3. The keyword _NOT_ excludes matching events.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="_stream:http_"
  (status=404 _OR_ status=500)
  _NOT_ status=200
```

STUDENT EXERCISE

Using a sourcetype you discovered in Lab 1, build a search that filters to it AND excludes one common/benign value you notice.

index=botsv2 earliest=0 sourcetype=access_combined NOT status=200

Lesson 5 — Searching Like a SOC Analyst

VOCABULARY CHECK — fill in the blank

1. The analyst search loop, in order: start _broad_ → identify useful fields → _narrow_ → summarize → investigate.
2. A(n) _search hypothesis_ is a plain-language guess to test, not an assumed fact.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 user=_admin_
| stats count by _host_
| sort - _count_
```

STUDENT EXERCISE

Write one search hypothesis in plain English about the BOTS v2 environment, then build SPL that actually tests it.

1. Search Hypothesis

I hypothesize that the BOTS v2 environment contains HTTP requests that resulted in errors, which may indicate failed requests or potentially suspicious activity.

2. SPL Search

index=botsv2 earliest=0 sourcetype=access_combined

| search status>=400

| stats count by status

| sort - count

Lesson 6 — Time Is Everything

VOCABULARY CHECK — fill in the blank

1. _earliest=0_ = search from the beginning of indexed time (training-data-only usage).
2. The _span_ parameter controls the bucket size in a timechart.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 earliest=_0_
| timechart span=_1h_ count
```

STUDENT EXERCISE

Run a timechart for one interesting sourcetype. Identify the busiest hour and explain what you'd check next before calling it suspicious.

1. SPL Search

index=botsv2 earliest=0 sourcetype=access_combined

| timechart span=1h count

2. Busiest hour: August 11, 2017, at 10:00 PM, with 4,793 events. (_time: 2017-08-11 22:00:00, Events: 4793)

Lesson 7 — Security Fields & Field Discovery

VOCABULARY CHECK — fill in the blank

1. The command _fieldsummary_ lists every field present in a result set along with counts and top values.
2. A good analyst does not memorize every field — they know how to _discover_ them.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="_<sourcetype>_"
| _| fieldsummary_
```

STUDENT EXERCISE

Run fieldsummary on two different sourcetypes. List three fields present in one but absent in the other.

The two sourcetypes used:

index=botsv2 earliest=0 sourcetype=access_combined

| fieldsummary

index=botsv2 earliest=0 sourcetype=stream:http

| fieldsummary

Three fields present in “access_combined” that are absent in “stream:http” are: action, clientip and eventtype

Lesson 8 — Windows Telemetry Investigation

VOCABULARY CHECK — fill in the blank

1. Windows telemetry can reveal: authentication, account activity, process creation, and _privilege_ activity.
2. High account activity is / is not (circle one) automatically suspicious.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="_<win_sourcetype>_"
| stats count by _user_
| sort - count
```

STUDENT EXERCISE

Identify one high-activity account. Build a table of its events with _time, host, user, and action (or the equivalent field you discovered).

```
index=botsv2 earliest=0 sourcetype="WinEventLog:Security" Account_Name="MERCURY$"
| table _time host Account_Name EventCode
| sort - _time
```

High-activity account: MERCURY$

Sourcetype: WinEventLog:Security

Fields displayed: _time, host, Account_Name, and EventCode

Event details:

- Time: August 29, 2017, at 7:11:36 PM (Splunk display time)

- Host: mercury.frothly.local (mercury)

- Account: MERCURY$

- EventCode: 4634

- Event type: Logoff

- Logon type: 3 (network logon)

- Security ID: NT AUTHORITY\SYSTEM

- Account domain: FROTHLY

- Logon ID: 0xA73FF13

- Audit status: Success

Observation: MERCURY$ appears to be a Windows computer account. Event ID 4634 indicates that a logon session was terminated. The log identifies a network logon session associated with the NT AUTHORITY\SYSTEM security context. I would correlate this event with the corresponding logon event using Logon ID 0xA73FF13 and inspect nearby events to determine what activity occurred before the session ended. This event alone does not indicate malicious activity.

Lesson 9 — Network Log Analysis

VOCABULARY CHECK — fill in the blank

1. The five core network questions: Who _initiated_? Where did it go? What _port_? How often? What protocol?

BUILD THE SPL — fill in the missing pieces

```
index=botsv2
| stats count by _src_ip_ dest_port
| sort - _count_
```

STUDENT EXERCISE

Identify one dest_ip that stands out. Investigate which host(s) and user(s) were involved.

Destination IP that stands out: 8.8.8.8

1. Why I chose this IP

In my Splunk results, 8.8.8.8 received 125,710 events from source IP 10.0.1.100 over destination port 53, which is commonly used for DNS traffic. Since 8.8.8.8 is Google's public DNS resolver, the traffic may be legitimate, but the high volume warrants further investigation.

2. Host involved

- Source IP: 10.0.1.100

- Source host: jupiter

- Destination IP: 8.8.8.8

- Destination port: 53 (DNS)

3. User involved

The Sysmon shows events associated with NT AUTHORITY\SYSTEM on wrk-abungst.frothly.local, involving Splunk-related processes such as splunkd.exe and splunk-powershell.exe.However, these events do not establish that this account generated the traffic from Jupiter to 8.8.8.8. The specific user responsible for the DNS traffic remains unconfirmed.

Lesson 10 — DNS Analysis

VOCABULARY CHECK — fill in the blank

1. The golden DNS rule: unfamiliar domain _≠_ malicious.
2. Regular, fixed-interval queries from one host may suggest _beaconing_.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="_<dns_sourcetype>_"
| stats count by _query_
| sort - count | head 20
```

STUDENT EXERCISE

Pick one non-obvious domain from your top-20 list. Investigate the host, frequency, and time window.

I selected uranus.frothly.local as the non-obvious domain for investigation. The DNS activity involved seven source IP addresses: 10.0.2.103, 10.0.2.110, 10.0.2.107, 10.0.2.101, 10.0.2.105, 10.0.2.108, and 10.0.1.120. Most of the traffic was directed to the DNS server at 10.0.1.100, while two events from 10.0.1.120 were directed to 192.175.48.1. The search returned 2,473 events, consisting of 1,239 DNS query events and 1,234 response events. The activity occurred between August 4, 2017, at 02:03:20 and August 29, 2017, at 19:10:49. Its frequency was uneven, with the highest observed hourly volume reaching 120 events during the hour beginning August 19, 2017, at 06:00. Additional spikes occurred on August 24 and August 26. This pattern warrants further investigation into the source hosts and their DNS activity, although the frequency alone does not establish that the domain or its traffic was malicious.

Lesson 11 — HTTP / Web Analysis

VOCABULARY CHECK — fill in the blank

1. A spike in _404_ responses from one source often suggests enumeration/scanning.
2. Scanning behavior looks like: many distinct paths, high _error_ rate, low repeat.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="_<web_sourcetype>_" status=_404_
| stats count by uri
| sort - count
```

STUDENT EXERCISE

Find one src_ip requesting an unusually high number of distinct URIs. Inspect what those paths are.

I investigated the source IP **203.149.168.246**, which generated 325 HTTP requests across 320 distinct URI paths in the BOTS v2 dataset. The requests targeted a Magento e-commerce website, including shopping cart, customer account, checkout, and JavaScript resource paths. The most frequently requested path in the inspected results was `/magento2/rest/default/V1/carts/mine/estimate-shipping-methods`, which appeared four times and returned HTTP 200 using the POST method. The host's activity was observed from **August 4, 2017, at 18:34:19 to 18:49:31**, lasting approximately 15 minutes and 12 seconds. The high ratio of distinct URI paths to total requests suggests that the host accessed a broad range of website resources in a short period. However, because many observed requests returned successful responses and the paths are consistent with Magento site functionality, further investigation is

needed before determining whether the activity represents malicious scanning or legitimate website use.

Lesson 12 — Stats Mastery — sort, dedup, top, eval

VOCABULARY CHECK — fill in the blank

1. _count_ counts total events; _dc_ (field) counts unique values of that field.
2. The _dedup_ command removes duplicate events, keeping one per unique field value.
3. The _eval_ command creates or transforms a field.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2
| stats _earliest_(_time) as first _latest_(_time) as last by host
| eval span=last-_first_
```

STUDENT EXERCISE

Using eval, calculate the activity time span (first-seen to last-seen) for one host of your choice.

I selected the host cassiopeia and used the earliest(), latest(), and eval commands to calculate its activity span. The host's first recorded event was on August 3, 2017, at 02:03:26, and its last was on August 29, 2017, at 19:13:05. This represents an elapsed time of approximately 641.16 hours, or 26.72 days.

Lesson 13 — Correlation — Thinking Across Data Sources

VOCABULARY CHECK — fill in the blank

1. Correlation chain, in order: IP → _Host_ → User → _Process_ → Network connection → Domain → Timestamp.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 dest_ip="_YOUR_DESTINATION_IP_"
| stats values(src_ip) values(_dest_port_) by dest_ip
```

STUDENT EXERCISE

Starting from the suspicious IP identified in Lab 2, build a cross-sourcetype timeline for the host involved, covering ±30 minutes around the activity.

The investigation covered August 31, 2017, from approximately 10:44 AM to 11:47 AM, representing a 30-minute window around the suspicious activity. The search returned 582 HTTP events involving destination IP 172.31.7.2. Source IP 212.83.203.115 generated 59 events at 11:10 AM and 232 events at 11:15 AM, while 144.79.191.211 generated 291 events at 10:50 AM. Only the stream:http sourcetype appeared in this time window, so the available results did not provide cross-sourcetype corroboration. Further investigation is needed to determine whether the activity was malicious.

Lesson 14 — Detections, Alerts & Tier 1 Triage

VOCABULARY CHECK — fill in the blank

1. A(n) _Detection_ is defined suspicious-behavior logic; a(n) _Alert_ is that logic scheduled to notify someone.
2. The 10-step triage workflow begins: Alert → _Validate the Alert_ → Identify affected asset → Identify user → ...

BUILD THE SPL — fill in the missing pieces

```
index=botsv2
| stats count by src_ip
| where count > _20_
```

STUDENT EXERCISE

Pick one src_ip from the detection output above. Run a full 10-question triage and decide escalate/close.

10-Question Triage Report

Selected source IP: 111.249.38.34

1. Is the alert valid?

Yes. The detection identified this IP because it generated 104,984 events, exceeding the threshold of 20 events.

2. Which host is affected?

The associated host is matar, communicating with destination IP 172.31.38.181.

3. Which user is involved?

No user was identified in the search results.

4. What is the time window?

August 19, 2017, from 20:52:23 to 22:04:33, approximately 72 minutes.

5. Is the activity isolated or repeated?

Repeated. The search returned 104,984 events during the observed time window.

6. What port and protocol are involved?

Destination port 25 over TCP, commonly associated with SMTP email traffic.

7. Which sourcetypes recorded the activity?

stream:ip, stream:smtp, and stream:tcp.

8. What behavior was observed?

The IP communicated with 172.31.38.181 over port 25. The large event count and SMTP-related traffic warrant further investigation, but the actual email activity has not yet been examined.

9. Is the activity benign, suspicious, or malicious?

Suspicious, pending further investigation. The high volume of activity and SMTP traffic are reasons to investigate, but they do not independently prove malicious behavior.

10. Should the case be escalated or closed?

Escalate for further investigation. Review the SMTP events to determine whether the activity involves unusual email volume, spam, unauthorized mail transmission, or other malicious behavior.

Lesson 15 — Timelines, Reporting & the Final Capstone

VOCABULARY CHECK — fill in the blank

1. The Tier 1 report field list includes: Alert, Time, Affected Host, User, Source IP, Destination IP, Observed Activity, Evidence, Assessment, Severity, Recommended Action, and _Escalation_.
2. Rule: write "the evidence indicates..." rather than "_this is definitely malware_."

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 host="_<host>_" earliest="_<start>_" latest="_<end>_"
| table _time sourcetype user src_ip dest_ip action process command
| sort _time_
```

STUDENT EXERCISE

Complete the reporting template for any prior lab using only evidence-based language.

The investigation was conducted in the botsv2 index on host jabbah for the time window of August 31, 2017, from 10:44:42 AM to 11:47:12 AM. The search returned 582 events. Of these, 498 contained both source and destination IP fields, while 84 were missing at least one of these fields. The results identified two groups of HTTP events, each containing 249 events. The first group originated from 144.79.191.211 and communicated with destination IP 172.31.7.2 between 10:50:49 AM and 10:53:18 AM. The second group originated from 212.83.203.115 and communicated with the same destination between 11:14:42 AM and 11:17:11 AM. These events were categorized as stream:http. The available evidence confirms HTTP activity between these IP addresses but does not establish that the activity was malicious. Further investigation of the 84 events with missing IP fields and inspection of the HTTP request details would be necessary to determine whether the observed activity was suspicious or associated with unauthorized behavior.

Lab Worksheets 1–10

Complete each lab using your own BOTS v2 environment. Fill in every blank in the guided steps before running the search, then answer the investigation questions in the space provided.

Lab 1 — Find Your Data

Scenario: You are a newly hired Tier 1 Analyst. Your first task is to understand what telemetry is available in the SIEM before investigating anything.

GUIDED STEPS — fill in the blanks

```
index=botsv2 earliest=0
| stats _count_

index=botsv2 earliest=0
| stats count by _sourcetype_
| sort - count

index=botsv2 earliest=0
| stats count by _host_
```

INVESTIGATION QUESTIONS

1. How much data exists in total?

The botsv2 index contains 68,870,348 events, according to the search results.

2. How many distinct sourcetypes are present?

There are 103 distinct sourcetypes in the index.

3. Which sourcetype contains the most events?

The sourcetype with the most events is mysql:server:stats, containing 13,579,885 events. It is followed by mysql:transaction:details with 13,574,451 events and Perfmon:Process with 11,868,094 events.

4. What categories of security telemetry appear to be available (Windows, network, web, DNS, etc.)?

The index contains database telemetry, including MySQL server statistics and transaction details; Windows system monitoring, including WinRegistry, WinHostMon, and Perfmon data; Windows security logs through WinEventLog:Security; network telemetry through stream:smtp, stream:ip, and stream:tcp; firewall traffic through pan:traffic; intrusion detection data through suricata; and Linux auditing through linux_audit and auditd. These sourcetypes indicate that the SIEM collects data from multiple systems and security-relevant sources.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Benign at this stage of the investigation. The inventory identified 68,870,348 events across 103 sourcetypes and 23 host entries. The host results show that cassiopeia contributes the most events, with 28,909,580, followed by MACLORY-AIR13 with 8,892,095 and mercury with 3,532,277. The dataset includes database, endpoint, network, firewall, intrusion detection, and audit telemetry. These findings describe the available data but do not independently demonstrate malicious activity.

Escalation decision:

No escalation at this stage. The searches were limited to inventorying the SIEM's data volume, sourcetypes, and hosts. No specific indicator of compromise or confirmed security incident was identified. Further investigation would be warranted if subsequent searches reveal suspicious activity.

Lab 2 — Investigate a Suspicious IP

Scenario: Threat intelligence reports a suspicious IP address. Your task is to determine whether systems in the environment communicated with it.

GUIDED STEPS — fill in the blanks

```
index=botsv2
| stats count by src_ip
| sort - count   (use this to pick a candidate IP)

index=botsv2 src_ip="_candidate_"

index=botsv2 src_ip="_candidate_"
| stats count by src_ip _dest_ip_
```

INVESTIGATION QUESTIONS

1. Is this an isolated event or repeated behavior? It is repeated behavior, with 249 HTTP events recorded for the source IP.
2. Which host and user are tied to this activity?The associated host is jabbah. The user field was empty in the results, so the specific user could not be identified from the available evidence.
3. What time window does the activity span? The activity occurred on August 31, 2017, from 11:14:42 to 11:17:11, lasting approximately 2 minutes and 29 seconds.
4. What port and protocol are involved? The traffic was directed to destination IP 172.31.7.2 on port 80, using HTTP, as indicated by the stream:http sourcetype.

Analyst conclusion (benign / suspicious / malicious, with evidence):

The source IP 212.83.203.115 generated 249 HTTP events to 172.31.7.2 over approximately 2 minutes and 29 seconds on August 31, 2017. The traffic targeted a Magento website, including shopping cart, customer login, checkout, and payment resources. Most observed requests returned HTTP 200, while a login-related request returned HTTP 302. The repeated activity warrants further investigation, but the available evidence is insufficient to confirm malicious behavior. The activity is therefore classified as suspicious pending further investigation.

Escalation decision:

Continue investigating and escalate for further review if additional evidence indicates malicious activity. Examine related sourcetypes, requested URI paths, HTTP methods, and surrounding events before making a final determination.

Lab 3 — Suspicious User Activity

Scenario: The SOC received a report that a user account may have been compromised.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_WinEventLog:Security_" user="_grace.hoppy_"
| table _time host user _Source_Network_Address_ action
| sort _time
```

INVESTIGATION QUESTIONS

1. Where did the user authenticate from?

The successful remote interactive logon for grace.hoppy originated from 10.0.4.1 and accessed mercury.frothly.local.

2. When did the activity occur?

The confirmed logon occurred on August 16, 2017, at 11:30:47 PM. The follow-up search covers activity through 11:40 PM.

3. Which systems were involved?

The confirmed destination system is mercury.frothly.local. The observed source addresses include 10.0.4.1 for grace.hoppy and 10.0.1.1 for separate service3 network-logon events.

4. Was there suspicious activity immediately after login?

The results show process creation and privilege-related events around the login time, including cmd.exe and explorer.exe. However, these can occur during normal Windows sessions. The available evidence does not establish that the activity was malicious.

5. Does the overall pattern look normal for this account?

The activity cannot yet be classified as normal or abnormal without comparing the source address, login time, and process activity against the account's expected behavior. The successful remote interactive logon warrants review, but it is not sufficient evidence of compromise.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Suspicious, pending further investigation. A successful remote interactive logon for grace.hoppy occurred on mercury.frothly.local from 10.0.4.1 on August 16, 2017, at 11:30:47 PM. The surrounding events include special-privilege assignment and process creation. These observations justify further review, but do not independently prove unauthorized access or malicious execution.

Escalation decision:

Escalate for further investigation if the source address or remote logon was unexpected, or if subsequent process activity is found to be unauthorized. Verify the relevant event details before declaring the account compromised.

Lab 4 — DNS Investigation

Scenario: A workstation is generating unusual DNS activity.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_stream:dns_"
| stats count by _query_
| sort - count
```

INVESTIGATION QUESTIONS

1. Which domain(s) stand out, and why?

hildegardsfarm.com stands out with 66,020 DNS events in the displayed results. This is considerably higher than the next entries, such as outlook-namnorth3.office365.com with 27,836 events. The unusually high volume warrants further investigation, although frequency alone does not establish maliciousness.

2. Which host queried it, and how often?

The host is jupiter. The largest group contains 32,570 events, from source IP 10.0.1.100 to destination 8.8.8.8. Other rows show smaller counts involving jupiter and other source or destination IPs.

3. Is the query timing regular/periodic, or irregular?

The five-minute time chart shows a substantial increase in DNS activity on August 26, 2017, beginning around 14:50. Several consecutive five-minute buckets contain hundreds or thousands of events, with counts varying between buckets. This indicates sustained, high-volume activity during that period, but the available chart does not establish a precise beaconing interval.

4. What corroborating evidence (network/HTTP) supports or weakens your hypothesis?

The current screenshot establishes high DNS query volume but does not show related HTTP requests, connections, or security alerts. Check for network activity involving the identified source host or IP before drawing a conclusion.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Classification: Suspicious, pending further investigation.

The DNS logs show a high volume of queries associated with hildegardsfarm.com, including a period of sustained activity on August 26, 2017. This behavior warrants investigation, but the available evidence is insufficient to confirm malicious activity.

Escalation decision:

Escalate for further investigation. Review the queried domain's resolved IP addresses, DNS response codes, related HTTP connections, and other activity from jupiter during the same time window. Do not label the domain malicious based on query volume alone.

Lab 5 — Build a Basic Detection

Scenario: You have been asked to create a basic detection for unusually high connection activity.

GUIDED STEPS — fill in the blanks

```
index=botsv2
| stats count by src_ip
| sort - count

index=botsv2
| stats count by src_ip
| where count > _10_
```

INVESTIGATION QUESTIONS

1. How many src_ips match at threshold > 10? 478 source IPs
2. How many match at > 20? 413 source IPs
3. How many match at > 50? 346 source IPs
4. Which threshold would you recommend, and why?

I would initially recommend greater than 20 events as a baseline. It is more selective than a threshold of 10, which may produce more routine network activity, while remaining more sensitive than a threshold of 50. The best threshold depends on normal network traffic and the number of false positives.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Classification: Suspicious, pending investigation.

The searches identified 478 source IPs with more than 10 events, 413 with more than 20 events, and 346 with more than 50 events. These results demonstrate that the detection identifies source IPs with relatively high event counts. However, event volume alone does not establish malicious behavior. Further investigation of destination IPs, ports, protocols, timing, and related security events is needed.

Escalation decision:

Escalate source IPs with unusually high counts for further investigation. Prioritize IPs with additional suspicious indicators, such as unusual destinations, unexpected ports, repeated connection patterns, or correlated security alerts. Do not classify an IP as malicious based solely on its event count.

Lab 6 — Brute-Force Authentication Investigation

Scenario: Multiple failed logins have been reported against one account.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_linux_secure_"
| stats count by user, _action_
| where _action_="failure" AND count > _5_
```

INVESTIGATION QUESTIONS

1. Do the failures come from a single source or many?

   Multiple source addresses are present, but the failures are heavily concentrated on a few sources. The failed logons came from multiple source-address values. There were 3,162 failed logon events from 10.0.1.100, 109 from 10.0.1.220, 54 with no recorded source address, 14 from 172.16.0.1, and 1 from 127.0.0.1. The largest concentration came from 10.0.1.100, which accounted for most of the 3,340 failed-logon events returned by the search.

2. Is a failure streak followed by a success?

   The logs for account MERCURYS and source address 10.0.1.100 showed repeated failed logons (EventCode 4625) interspersed with successful logons (EventCode 4624) on August 4, 2017. This pattern warrants further investigation, although the available results do not establish that a specific uninterrupted streak of failures was immediately followed by a success or that the successful logons resulted from an attack.

3. What MITRE ATT&CK technique would you consider mapping this to, and why only after reviewing evidence?

   The activity could be investigated under MITRE ATT&CK T1110: Brute Force, which covers attempts to gain access by repeatedly trying credentials. The large number of failed logons and the presence of successful logons justify further review. However, this technique should remain a candidate mapping until the event sequence, account activity, source-system identity, and expected behavior are examined. The current evidence does not confirm deliberate credential guessing.

Analyst conclusion (benign / suspicious / malicious, with evidence):

The activity is classified as suspicious. The investigation identified 3,162 failed logon events associated with 10.0.1.100, and the event review showed both failed and successful logons for MERCURYS. These findings warrant further investigation, but they do not independently prove malicious intent or confirm a brute-force attack.

Escalation decision:

Escalate for further investigation by the security analyst or incident response team. Validate the source address, determine whether the authentication activity is expected, and examine successful logons occurring after repeated failures. Do not declare a confirmed security incident until additional evidence supports that conclusion.

Lab 7 — Suspicious PowerShell Activity

Scenario: An analyst suspects PowerShell misuse on a host.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_XmlWinEventLog:Microsoft-Windows-Sysmon/Operational_"
| search _Image_="*powershell*"
```

INVESTIGATION QUESTIONS

1. Is there evidence of an encoded command or download cradle?

   Yes. Multiple Sysmon process-creation events show PowerShell commands using -enc or -EncodedCommand to execute Base64-encoded scripts, often with hidden windows. Some commands use System.Net.WebClient to download content from the remote server 45.77.65.211 over HTTPS port 443 and execute it using IEX (Invoke-Expression). Other commands retrieve encoded data from Windows registry keys before decoding and executing it. These behaviors are strong indicators of suspicious, potentially malicious PowerShell activity.

2. What is the parent process of the PowerShell execution?

   The observed parent processes include taskeng.exe, WmiPrvSE.exe, powershell.exe, and eventvwr.exe. The events occur across the hosts wrk-klagerf, wrk-btun, and venus. The use of multiple parent processes suggests different execution paths that require further investigation, particularly the PowerShell activity launched through Windows Management Instrumentation and Event Viewer.

3. Is this use consistent with normal admin activity, or does it look abnormal? Why?

   The activity is abnormal and highly suspicious because it combines encoded commands, hidden PowerShell execution, registry-based payload retrieval, remote downloads, and execution through IEX. These techniques can be used to conceal malicious scripts and execute payloads without directly writing them to disk. The involvement of multiple hosts and different parent processes further increases concern that this may be part of a broader compromise rather than routine administrative activity.

Analyst conclusion (benign / suspicious / malicious, with evidence):

- Malicious
The collected evidence strongly indicates malicious PowerShell execution across multiple hosts. The encoded scripts, hidden execution, registry-based retrieval, and remote payload download and execution are inconsistent with ordinary administrative activity. Although the precise payload and full impact have not yet been established, the observed behaviors justify treating this activity as malicious pending further incident-response analysis.

Escalation decision:

Escalate immediately to the incident-response team for investigation. Preserve the relevant logs, investigate communications with 45.77.65.211, analyze the encoded payloads, examine registry changes and potential persistence mechanisms, and determine whether additional hosts were affected.

Lab 8 — Network Reconnaissance

Scenario: Possible port or host scanning is suspected.

GUIDED STEPS — fill in the blanks

```
index=botsv2
| stats dc(dest_port) by _src_ip_
| sort - dc(dest_port)

index=botsv2
| stats dc(_dest_port_) by src_ip
| sort - dc(_dest_port_)
```

INVESTIGATION QUESTIONS

1. Which src_ip touches the widest breadth of ports or destinations?

   10.0.1.120 has the widest destination-port breadth, contacting 5,769 distinct destination ports across 24 distinct destination IPs. For destination-host breadth, 10.0.2.107 leads with 566 distinct destination IPs, followed by 10.0.2.103 with 445 and 10.0.2.109 with 375.

2. Does this pattern look like scanning rather than normal usage? Why?

   The patterns are suspicious and may indicate network reconnaissance. The source 10.0.1.120 contacted thousands of distinct ports, while 10.0.2.107 contacted hundreds of distinct destination IPs. Such broad activity can be consistent with port scanning or host discovery. However, the distinct counts alone do not prove scanning because legitimate monitoring or network-management activity can also contact many ports and hosts. Reviewing the connection timeline and destination distribution would help determine whether the activity is abnormal.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Classification: Suspicious, pending further investigation.

The results show unusually broad network activity. Source 10.0.1.120 contacted 5,769 distinct destination ports, while 10.0.2.107 contacted 566 distinct destination IPs. These patterns warrant investigation for possible port scanning or host discovery, but additional evidence is needed to confirm malicious intent.

Escalation decision:

Investigate the leading source IPs, especially 10.0.1.120 and 10.0.2.107. Review their connection timestamps, destination addresses, port distributions, and whether the activity matches authorized network-monitoring or administrative tasks. Escalate if the timeline confirms rapid probing or other indicators of unauthorized reconnaissance.

Lab 9 — Web Attack Investigation

Scenario: Possible web exploitation attempt against an internet-facing system.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_stream:http_"
| stats count by _http_method_ uri
| sort - count
```

INVESTIGATION QUESTIONS

1. Are there unusual HTTP methods or status-code patterns?

   The HTTP status summary shows 8,470 404 responses, 4,314 412 responses, and smaller numbers of 403, 405, and other responses. The detailed results show repeated 404 responses for requests to Magento's coupon API using different coupon codes. This pattern may indicate automated coupon enumeration rather than ordinary browsing.

2. Do any requested URIs look like exploitation attempts (e.g., admin panels, config files)?

   Yes. Requests to /magento2/rest/default/V1/carts/mine/coupons/ with numerous different coupon codes are suspicious because they may represent attempts to identify valid promotional codes. The displayed requests returned 404, so there is no evidence in this output that the attempts succeeded.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Analyst conclusion: Suspicious, with evidence of possible automated coupon enumeration.

Multiple source IPs requested different coupon codes through the Magento REST API, and the displayed requests returned 404 responses. This pattern is consistent with automated probing of the coupon endpoint, but successful exploitation has not been demonstrated.

Escalation decision:

Escalate for further investigation if this activity is unauthorized. Correlate the requests with source IPs, timestamps, HTTP methods, and successful responses to determine whether any valid coupons were discovered or abused.

Lab 10 — Multi-Stage Timeline & Full Triage

Scenario: Combine findings from Labs 2, 4, and 8 into a single cross-sourcetype timeline and run a full Tier 1 triage.

GUIDED STEPS — fill in the blanks

```
index=botsv2 host="_jupiter_" earliest="_08/26/2017:14:45:00_"
latest="_08/26/2017:15:30:00_"
| table _time sourcetype user src_ip dest_ip action
| sort _0__time_
```

INVESTIGATION QUESTIONS

1. What story does the chronological sequence tell?

   The timeline shows a high volume of network events across several sourcetypes during the selected 45-minute period. The results include Suricata alerts and network traffic records from TCP, IP, and DNS sources. Multiple internal addresses, including 10.0.1.100 and 10.0.2.189, appear alongside external destinations such as 45.77.65.211 and 8.8.8.8. This indicates activity across multiple network protocols that warrants correlation. However, the screenshot shows only the beginning of the timeline, and the displayed events alone do not establish a complete attack sequence or prove that the different activities are related.

2. What is your final severity assessment?

   Severity: Medium, pending further investigation. The volume of events and the presence of multiple internal and external addresses warrant further review, especially when considered alongside the unusual DNS and network breadth findings from the earlier labs. However, event volume and the presence of external destinations are not sufficient to confirm a compromise. The severity should be revised if the correlated events reveal confirmed malicious activity or impact on a host.

3. Escalate or close — and why?

Analyst conclusion (benign / suspicious / malicious, with evidence):

The cross-sourcetype search returned 15,437 events involving Suricata, TCP, IP, and DNS data during the selected period on August 26, 2017. The results show network activity involving multiple internal hosts and external destinations, warranting further investigation in conjunction with findings from Labs 2, 4, and 8. The available evidence does not yet confirm a unified multi-stage attack. The case should remain open at medium preliminary severity while the events are correlated and suspicious activity is validated.

Escalation decision:

Decision: Keep the case open for further investigation. Review the Suricata alerts and correlate them with DNS queries and TCP/IP traffic using timestamps, source IPs, destination IPs, and any available action or alert fields. Determine whether the events connect the suspicious activity identified in earlier labs to the same hosts or time period. Escalate to incident response if the correlation reveals malicious connections, confirmed reconnaissance, or evidence of compromise. Do not close the case based on the current timeline alone.